Zain Dana Harper

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AI Evaluation and Test Engineer

Summary

I build tests that tell a working AI system from a broken one, and I fix other people's software: maintainers of 34 open-source projects have merged 41 of my pull requests. About thirteen years of programming, and eleven years of commercial tree care scoping jobs and coordinating crews.

Selected Results

- Merged 41 pull requests in 34 other people's projects, 3 of them list entries, counted November 14, 2026. DeepEval #2822 fixed a crash on unhashable tool outputs, with regression tests. Pydantic AI #6205 kept a file-download setting through message round trips. Datasette #2815 turned a server error into a client error, with tests. harperz9.github.io/portfolio.html
- Shipped Flywheel 1.9.0, Articulate 0.18.0 and Accountable Surface 0.3.1 on PyPI, BuildLang 1.15.0 on crates.io, and Phantom 1.1.1 and CMMNS 3.10.0 as GitHub releases. Versions read November 14, 2026.
- Released Elder ENB, a Skyrim lighting mod: 1,004,693 downloads (179,218 unique) and 2,418 endorsements as of November 13, 2026. nexusmods.com/skyrimspecialedition/mods/117327
- Published Who Knew First, an AI-assisted investigation of nine reported AI-agent incidents; 30 research findings, 12 of them nulls; 13 DOI records. Not peer reviewed. harperz9.github.io/published-work.html

Experience

Arboriculture and Field Operations | Legendary Tree

Seattle area | April 25, 2015 to June 2, 2026

- Turned site assessments into estimates and job scopes; coordinated clients, crews and vendors on access, timing and hazards; watched rigging and clearances from the ground.

Technical Networking Support | Stream/Convergys | 2014 to 2015

- Resolved Xbox connectivity problems on a Microsoft product contract in Wilsonville, Oregon, by phone and chat, across TCP/IP, DNS, NAT, firewalls and routers; documented the steps for support handoffs.

I have not yet held a paid software role. The software on this page is independent work.

Projects

- Measured which guards stop destructive tool calls from a steered agent, on mocked tools: a request-versus-action check caught 64 of 66 and a schema check caught 1 of 66. Merged, not yet released.
- Built Flywheel's behavior check, which flagged 20 of 20 planted changes with 0 false alarms in 50 unchanged pairs. A separate accuracy comparison showed no measured gain, and I retired that benchmark.
- Built 324 terminal-state fixtures and 337 component cases that tell a failed answer from a provider failure; ran a pinned METR count_odds task through the Inspect bridge with three deterministic controls.
- Wrote guides, API documentation and release notes as a freelance technical writer, from 2017.

I write with coding agents and review, test and release the work myself.

Education

High School Diploma | Wilsonville High School, Oregon | 2013

Credentials, 2026: Google DeepMind skill badge, 4 Microsoft Applied Skills, 6 Qualys specialist exams.

Skills

Python, Rust, C++, Dart/Flutter, SQL, pytest, GitHub Actions, Linux, Windows, MCP; evaluation design, planted faults and controls, red-team test cases.